

DANIEL OPEYEMI

UK | +44 7867 066637 | danielopeyemi840@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL PROFILE

Machine Learning Engineer with 3 years' experience across AI/ML, data engineering and telecommunications, currently studying MSc Machine Learning & Deep Learning at the University of Strathclyde. Track record of delivering measurable results, including a banking intent-classification model achieving over 95% accuracy and a cloud-removal GAN pipeline that raised reconstruction PSNR from 16.94 to 21.33 dB. Skilled in Python, TensorFlow, PyTorch, RAG and Azure, with a strong grounding in building and deploying production-ready AI/ML solutions.

KEY SKILLS

Languages: Python, SQL, JavaScript, Java, C

Generative AI & LLMs: LLM architectures, prompt engineering, retrieval-augmented generation, model fine-tuning and optimization (transfer learning, LoRA and QLoRA), Hugging Face Transformers

Machine Learning & Data Science: Supervised classification, NLP, deep learning and CNNs, GANs, transformers, RAG and embeddings, prompt engineering, model benchmarking, TensorFlow, PyTorch, scikit-learn, Pandas/NumPy

RAG & Retrieval: LangChain, LlamaIndex, embeddings and cosine similarity, vector databases, semantic and hybrid search

Cloud & Tools: Azure, AWS, Docker, CI/CD, Git, Linux, Azure Speech, REST API, model monitoring and deployment.

Data Engineering: ETL pipelines, PostgreSQL, data modelling, automated data quality control, metadata & lineage, large-scale datasets

Professional: Agile and Scrum cross-functional delivery, code reviews, experiment design and metric-driven evaluation, stakeholder communication

PROFESSIONAL EXPERIENCE

AI Research Engineer (MSc Dissertation) | University of Strathclyde, Glasgow, UK 2026 – Present

Automated SAR-optical data pipeline and improved GAN architecture for cloud removal from Sentinel-2 imagery

- Built an automated data pipeline integrating the Copernicus API, patch-level cloud segmentation and matched-pair assembly, expanding the 15-channel SAR-optical training dataset from 28,617 to 58,458 triplets under automated quality control across ten Sentinel-2 tiles.
- Diagnosed a silent Sentinel-2 radiometric offset affecting post-2022 imagery and implemented a targeted correction step to prevent corrupted training data.
- Traced reconstruction artefacts to transposed-convolution up-sampling, batch normalisation test-time statistics and image normalisation, improving PSNR from 16.94 to 21.33 dB, SSIM from 0.630 to 0.794, and SCC from 0.290 to 0.367.
- Trained and evaluated GAN models on the ARCHIE-WeSt HPC cluster using SLURM for GPU workloads, running 15+ controlled experiments using PSNR, SSIM and SCC.

AI & ML Engineer | Wema Bank (Digital Banking) Aug 2025 – Dec 2025

- Worked within a team of five to develop an end-to-end AI/ML conversational banking assistant, building Python/FastAPI inference APIs, containerising services with Docker, and deploying to a live Azure endpoint through CI/CD.
- Trained a logistic regression intent-classification model on 11,035 samples distinguishing card and transfer issue queries, achieving over 95% accuracy, and integrated it with a RAG pipeline using OpenAI text embeddings and cosine similarity for banking-content retrieval.
- Implemented confidence-based fallback routing, directing predictions below a 0.60 confidence threshold to an unsupported-request flow to prevent unreliable answers.

Network Performance Engineer (Large-Scale Telecoms Data) | Huawei Dec 2023 – Jul 2025

- Built Python/PostgreSQL ETL pipelines and automated daily KPI reporting, reducing reporting turnaround from 3–4 hours of manual work to under 30 minutes across recurring optimisation reports.
- Analysed large-scale 2G to 5G network telemetry with Python and Pandas to support reporting and capacity analytics, turning raw performance data into measurable insight on degradation.
- Collaborated with optimisation and field teams to prioritise sites and validate fixes against KPI targets, keeping network efficiency KPIs above 97% and 2G packet loss below 0.5%.

Software Development Engineer (IT Intern) | IHS Towers Mar 2022 – Dec 2022

- Developed RESTful APIs in Python and SQL for authentication, data retrieval and workflow automation, working across networking teams to integrate services and improve system stability.

EDUCATION

MSc Machine Learning & Deep Learning | University of Strathclyde, Glasgow, UK

Jan 2026 – Jan 2027

- Coursework: statistical learning, deep learning theory, transformer architecture, sequence modelling, representation learning, generative models, information retrieval, NLP.

BEng Computer Engineering – First Class Honours | Covenant University

2018 – 2023

- Relevant modules: data structures & algorithms, software engineering, communication systems, digital signal processing, AI/ML.

SELECTED PROJECTS

Data Engineering & Web Scraping | Python · PostgreSQL · Playwright · Selenium

- Built ETL pipelines over PostgreSQL and REST APIs and automated extraction with an asynchronous Playwright scraper, BeautifulSoup and Selenium, feeding predictive models built with Pandas and scikit-learn.

Brain Tumour Classification, Medical App | Python · TensorFlow · Flask

- Trained a CNN with transfer learning, reaching 99% test accuracy on MRI brain-scan classification, and served it through a Flask inference API integrated into a Flutter mobile app.

CERTIFICATIONS

AI & ML Engineer (Dbrown Academy / Wema Bank) · Data Analytics & Data Science (IBM) · Python & Django (Udemy)